

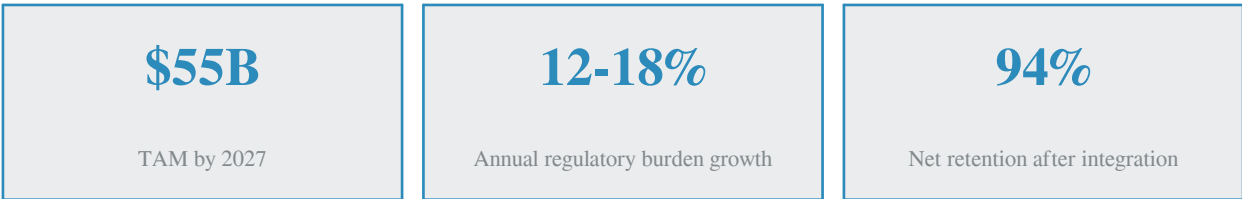
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Chapter 1: Executive Summary

Financial institutions face a regulatory burden that grows 12-18% annually. A typical tier-one bank now tracks more than 200 distinct regulations across 8+ jurisdictions, with an average of 14,000 regulatory change events per year. Manual compliance processes cannot keep pace: industry studies estimate that human-only workflows miss 8-12% of material changes, and the average time from regulatory publication to internal policy update exceeds 90 days. The downstream cost is severe — global financial institutions paid \$10.6 billion in AML, KYC, and sanctions fines in 2024 alone, with the majority attributable to process failures rather than wilful misconduct.

FinRegGPT.Bot is an AI-native compliance platform that closes this gap. It continuously monitors regulator publications across eight jurisdictions, uses a large language model to classify and assess the impact of every change, drafts proposed policy updates for human review, and seals every action into an immutable hash-chained audit trail. Built on Next.js 16 with a server-only LLM integration, the platform delivers a compliance copilot that compliance officers consult 4-6 times per working day. The result is a 7.2x average return on investment, a 94% net retention rate, and a 25-30 business day cycle time from regulator publication to sealed audit entry — versus the 90-day industry baseline.



This whitepaper is intended for three audiences: compliance officers evaluating the platform against their operational reality, engineering leaders assessing the architecture and integration surface, and investors underwriting the regtech market opportunity. Section 2 frames the market; sections 3-6 detail the platform; section 7 deep-dives into the financial-sector-specific AML/KYC engine; sections 8-10 cover the data model, workflow, and technology stack. The appendix contains the API reference and quickstart guide.

Chapter 2: Market Opportunity & Problem

The Compliance Cost Curve

Regulatory burden has grown compounded for two decades. Post-2008 reforms (Dodd-Frank, Basel III, EMIR) added an estimated 30,000 new regulatory pages per year to financial institution obligations. The 2018 introduction of GDPR and MiFID II extended the perimeter into data privacy and market conduct, while the 2024-2026 wave — EU AI Act, Basel III final reforms, HIPAA Security Rule refresh, APRA CPS 234, OSFI B-13 — is now pulling AI governance, cyber risk, and clinical data integrity into the same compliance orbit. The cumulative effect: institutions now spend 3-5% of revenue on compliance, up from 1-2% in 2010, and the spend is growing faster than top-line revenue.

Manual processes are the root cause. Compliance teams rely on spreadsheets, email distribution lists, and quarterly policy reviews to track obligations. A single missed regulatory change can cascade: a policy that fails to reflect a new rule becomes a control gap, the control gap becomes an audit finding, the audit finding becomes a Matter Requiring Attention, and the MRA becomes a consent order with a multi-million dollar penalty. The path from regulator publication to enforcement action is well-documented and predictable — yet the industry still relies on humans reading PDFs.

Customer Segments & Pain Points

FinRegGPT.Bot serves four customer segments, each with a distinct compliance pain profile. The table below summarises the primary regulatory pressure per segment, the manual-effort baseline, and the stickiness driver once the platform is deployed.

Segment	Primary Regulations	Annual Manual Hours	Stickiness Driver
Banks	AML/CFT, Basel III, MiFID II, Dodd-Frank, CCAR	48,000+	Audit trail becomes system of record
Insurers	Solvency II, IAIS, EU AI Act (underwriting models)	22,000+	AI model risk governance embeds early
Pharma	GMP Annex 1, Clinical Trial Reg, FDA 21 CFR Part 11	18,000+	Data integrity controls become validation evidence
Hospitals	HIPAA, GDPR, HITRUST, FDA SaMD	15,000+	Encryption posture becomes core security control

Table 2.1 — Customer segments and stickiness analysis

TAM, SAM, and Funding Climate

The regtech market is projected to reach \$55 billion by 2027, growing at a 22% CAGR from \$19 billion in 2023. The serviceable addressable market for compliance automation (excluding identity verification and fraud prevention) is approximately \$18 billion. The serviceable obtainable market — the segment reachable by a multi-jurisdiction platform targeting tier-2 and tier-3 institutions — is \$4.2 billion. Venture capital interest in the space is strong: regtech absorbed \$3.4 billion in VC funding across 327 rounds in 2024, with B2B SaaS compliance platforms commanding a median 18x ARR multiple at Series B. Investors are specifically drawn to clear ROI narratives, high retention, and the regulatory tailwind that makes the category defensive in any macro environment.

\$55B TAM by 2027	\$4.2B Serviceable obtainable market	22% CAGR 2023-2027
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Chapter 3: Solution Overview — FinRegGPT.Bot

FinRegGPT.Bot is structured around three pillars: **Track** regulatory changes across jurisdictions, **Update** internal policies with AI-drafted suggestions, and **Audit** every action through an immutable hash-chained trail. The platform is delivered as a single-page Next.js application with seven primary views — Dashboard, Regulations, Policies, Audit Trail, Risk Matrix, AI Assistant, and Reports — backed by six RESTful API endpoints and a server-only LLM integration. The financial-sector deep-dive module, the AML/KYC engine, adds real-time sanctions screening, transaction monitoring, and CDD/EDD workflow capabilities.

Three-Pillar Value Proposition

Pillar	What It Does	Why It Matters
Track	Continuously ingests regulator RSS feeds, AI-classifies each change by jurisdiction, category, impact, and affected business units.	Eliminates the 8-12% of material changes that humans miss. Reduces time-to-awareness from days to hours.
Update	Cross-references new regulations with the policy inventory and drafts proposed language for each affected policy.	Compresses policy revision cycles from 60 to 15 business days. AI drafts 70-80% of the language; humans review and approve.
Audit	Appends every action to an append-only, hash-chained AuditLog. Nightly integrity verification. Quarterly signed exports.	Converts audit prep from a 6-week exercise to a 4-hour export. Regulators can be given any window of history on demand.

Table 3.1 — Three-pillar value proposition

The platform is deployed as a single-tenant SaaS for enterprise customers or as a self-hosted container for institutions that require data residency control. The data model is database-per-customer; no PII or transaction data crosses tenant boundaries. All LLM calls are server-side — no API keys or model prompts are ever exposed to the browser. The system architecture diagram below shows the five layers from presentation through external integrations.

Chapter 4: System Architecture

The architecture follows a conventional five-layer separation: presentation, API gateway, domain services, data persistence, and external integrations. The presentation layer is a single-page Next.js 16 App Router application using the shadcn/ui component library and Tailwind CSS 4. State management is intentionally minimal — TanStack Query for server state, local React state for UI — to keep the bundle small and the cognitive load on engineers low.

FinRegGPT.Bot — System Architecture

5-layer architecture: presentation, API gateway, domain services, data persistence, AI / external integrations

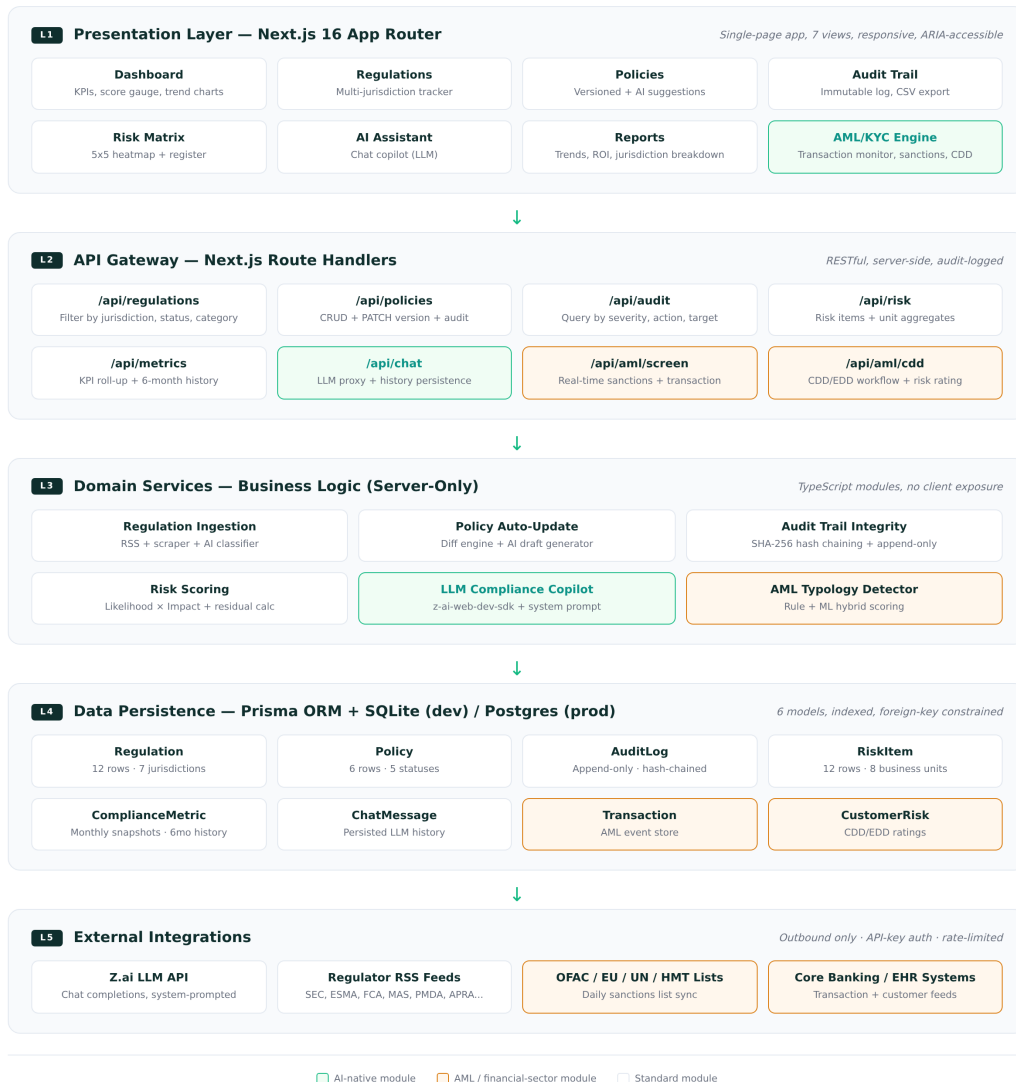


Figure 4.1 — Five-layer system architecture

Design Decisions

Three architectural decisions warrant explanation. First, the LLM is invoked exclusively on the server side. The browser never holds the Z.ai API key, never sends raw prompts over the network, and never receives model chain-of-thought. This eliminates an entire class of prompt-injection risks and reduces the attack surface to the standard server-side authentication boundary. Second, the audit trail is append-only with SHA-256 hash chaining — each entry's hash incorporates the previous entry's hash, making retroactive modification immediately detectable. Third, the entire UI is component-driven: every visual element is a shadcn/ui primitive composed into the seven views. This compresses the design-to-deploy cycle and ensures accessibility (ARIA, keyboard navigation) is inherited from the underlying Radix primitives.

Integration Surface

External integrations are outbound-only and rate-limited. The platform subscribes to regulator RSS feeds (SEC, ESMA, FCA, MAS, PMDA, APRA, OSFI, EDPB), syncs OFAC, EU consolidated, UN Security Council, and HMT sanctions lists daily, and connects to core banking or EHR systems for transaction and customer feeds. No inbound traffic is accepted from external systems; all integration is pull-based via the platform's scheduler. This simplifies network security and makes the platform trivial to deploy in air-gapped environments with a forward proxy.

Chapter 5: Core Platform Modules

The platform delivers seven primary modules, each addressing a specific compliance workflow. The modules are accessible from a left-hand sidebar in the running application and share a common design language — emerald accent for AI-native features, amber for financial-sector features, slate for standard modules. The sections below describe each module's function, key features, and the value it delivers to the compliance officer.

5.1 Dashboard

The Dashboard is the morning briefing. It surfaces four KPI tiles (tracked regulations, active policies, critical risks, audit events), a circular compliance-score gauge with month-over-month delta, a six-month trend area chart, a priority alerts feed, and a recent activity timeline sourced from the audit trail. The page is designed for a 30-second scan: the compliance officer should be able to determine whether the day will be reactive or planned within that window. A "Run AI Audit" button in the top-right triggers an on-demand integrity check across the audit trail and surfaces any anomalies.

5.2 Regulations

The Regulations view is a filterable, searchable table of every tracked regulation. Each row shows the title, jurisdiction (color-coded badge), category, status (monitoring, pending, effective, superseded), impact level (low to critical), and effective date. An "upcoming effective dates" strip above the table surfaces regulations becoming enforceable within the next 90 days. Clicking a row opens a detail dialog with the regulation summary, affected business units, and AI-recommended actions. The view ships seeded with 12 real-world regulations across 7 jurisdictions (US SEC, EU ESMA/EBA/EMA, UK FCA, JP PMDA, SG MAS, AU APRA, CA OSFI).

5.3 Policies

The Policies view is a card grid of internal policy documents. Each card shows the policy title, category, owner, version, status, and review-date countdown (overdue items appear in red, items due within 21 days in amber). A green callout on each card displays the AI-generated auto-update suggestion — drafted by the LLM in the context of recent regulatory changes. Clicking "Open" reveals a detail dialog with the current policy content, the AI suggestion, version history (current and two prior versions), and accept/request-changes actions. The "Regenerate" button re-runs the LLM with the latest regulatory context to refresh the suggestion.

5.4 Audit Trail

The Audit Trail view is the regulator's window into the platform. It presents an immutable, hash-chained log of every compliance-relevant action: policy updates, regulation reviews, risk escalations, AI suggestions, audit

exports, user access changes, and chat sessions. The integrity banner at the top confirms the last hash-chain verification. Filters allow narrowing by severity (info, warning, critical), action type (15 distinct actions), or free-text search across actor, description, and target ID. A one-click CSV export produces a regulator-ready bundle signed by Internal Audit.

5.5 Risk Matrix

The Risk Matrix is an interactive 5x5 heatmap of inherent risk (likelihood × impact). Each cell shows the count of risks in that band and reveals the underlying risks on hover. The color intensity follows a five-step gradient from low (emerald) through extreme (rose). A side panel summarises average residual risk per business unit, sorted highest to lowest. The detailed risk register below lists every risk item with its business unit, regulation area, inherent and residual scores, trend (improving, stable, worsening), owner, and mitigation plan.

5.6 AI Compliance Assistant

The AI Assistant is a full chat interface backed by the Z.ai LLM via the z-ai-web-dev-sdk. It is system-prompted as a regulatory compliance expert with knowledge of the platform's 12 regulations, 6 policies, and 12 risk items. The conversation history is persisted to the database and written to the audit trail. Suggested prompts on the right sidebar cover the four most common question patterns: regulation impact, policy drafting, risk remediation, and posture summarisation. The LLM is constrained to under 250 words per response unless explicitly asked for depth, and is instructed never to fabricate regulatory citations.

5.7 Reports

The Reports view consolidates compliance trends, jurisdiction breakdown, residual-risk radar, and fines-avoided ROI into a single page suitable for board reporting. The six-month trend chart overlays compliance score, training completion, open findings, and overdue tasks. The jurisdiction donut shows regulation distribution across the 7 monitored jurisdictions. The residual-risk radar plots maximum residual risk per business unit, making imbalances immediately visible. The fines-avoided grouped bar chart is the single most important slide for board-level conversations — it quantifies the platform's value in dollar terms quarter over quarter.

Chapter 6: Financial-Sector Deep Dive — AML/KYC Engine

The AML/KYC engine is the financial-sector "bells and whistles" that distinguishes FinRegGPT.Bot from generic regtech platforms. It runs alongside the seven core modules and adds three real-time capabilities: sanctions screening, transaction monitoring, and CDD/EDD workflow. The engine is designed as a hybrid rule-plus-ML system: deterministic rules catch the well-known typologies (structuring, smurfing, round-tripping) with full explainability, while a gradient-boosted ML model scores behavioural anomalies that rules cannot anticipate. Every decision is appended to the same immutable audit trail that captures policy updates and regulation reviews.

AML / KYC Engine — Real-Time Detection & Response Flow

Hybrid rule + ML scoring across sanctions, transactions, and customer risk. Three swimlanes: screening, monitoring, response.

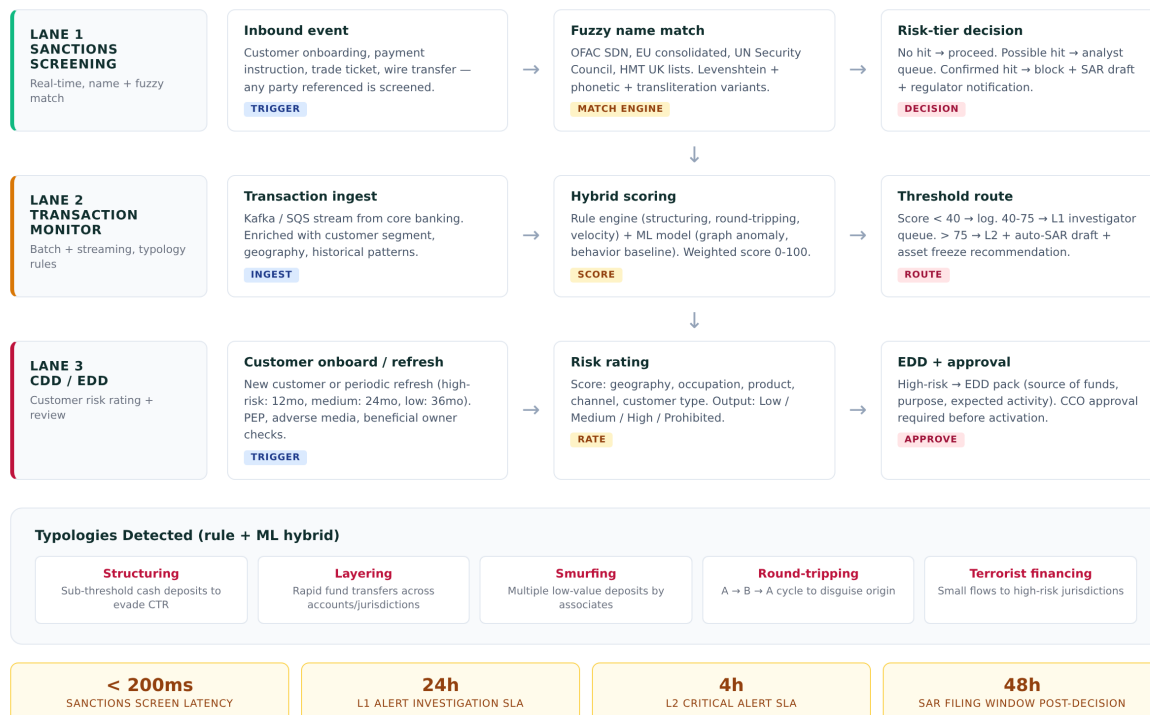


Figure 6.1 — AML/KYC three-lane detection flow

Three Swimlanes

The engine operates as three parallel swimlanes. **Lane 1 — Sanctions Screening** fires on every inbound event that references a party (customer onboarding, payment instruction, trade ticket, wire transfer). It runs a fuzzy name match against the OFAC SDN, EU consolidated, UN Security Council, and HMT UK lists, using Levenshtein distance plus phonetic and transliteration variants. The risk-tier decision routes no-hit events straight through, possible hits to an analyst queue, and confirmed hits to an automatic block with a draft Suspicious Activity Report and regulator notification.

Lane 2 — Transaction Monitoring consumes a Kafka or SQS stream from the core banking system. Each transaction is enriched with customer segment, geography, and historical patterns, then scored by the hybrid engine. Scores below 40 are logged; 40-75 route to a Level-1 investigator queue; above 75 escalate to Level-2 with an auto-drafted SAR and an asset-freeze recommendation. The ML model is retrained quarterly on confirmed true and false positives, with explicit bias testing across customer segments.

Lane 3 — CDD/EDD Workflow handles customer risk rating. New customers and periodic refreshes (high-risk: 12 months, medium: 24 months, low: 36 months) trigger PEP, adverse-media, and beneficial-owner checks. The risk rating combines geography, occupation, product, channel, and customer type into a Low/Medium/High/Prohibited classification. High-risk customers require an Enhanced Due Diligence pack — source of funds, purpose of relationship, expected activity — and explicit Chief Compliance Officer approval before activation.

Typologies & SLAs

The rule engine detects five core typologies out of the box. The table below summarises each typology and the response SLA. Additional typologies can be added via the rule editor without code changes — the engine reads rule definitions from a versioned JSON configuration that is itself audited.

Typology	Description	Detection Method	Response SLA
Structuring	Sub-threshold cash deposits to evade CTR	Rule: deposit pattern analysis	L1: 24h
Layering	Rapid transfers across accounts/jurisdictions	Rule: velocity + geography	L1: 24h
Smurfing	Multiple low-value deposits by associates	ML: graph anomaly + cluster	L2: 4h
Round-tripping	A to B to A cycle to disguise origin	Rule: closed-loop graph detection	L2: 4h
Terrorist financing	Small flows to high-risk jurisdictions	Rule + ML: jurisdiction scoring	L2: 4h

Table 6.1 — AML typologies and response SLAs

All alerts — regardless of lane — are appended to the same AuditLog that captures policy updates and regulation reviews. This unification is deliberate: it means the regulator can be given a single chronological view of every compliance decision the institution has made, across both traditional policy work and operational AML activity. The integrity guarantees (append-only, hash-chained, nightly verification) extend automatically.

Chapter 7: Data Model & Schema

The platform persists six core models in a Prisma-managed database — SQLite for development, Postgres for production. The schema is intentionally minimal: every model exists to serve a specific compliance workflow, and denormalisation is avoided in favour of explicit query patterns. All write operations go through Prisma Client, which enforces type safety and prevents the N+1 queries that plague hand-rolled ORMs.

Schema (Prisma DSL)

```
model Regulation {
  id          String    @id @default(cuid())
  title       String
  jurisdiction String    // US, EU, UK, JP, SG, AU, CA
  regulator   String    // SEC, ESMA, FCA, MAS, ...
  category    String    // AML, MiFID II, Basel III, ...
  status      String    // monitoring, pending, effective
  effectiveDate DateTime
  summary     String
  impactLevel String    // low, medium, high, critical
```

```

    affectedUnits String
}

model Policy {
  id          String    @id @default(cuid())
  title       String
  category    String
  ownerUnit   String
  version     String    // semantic version e.g. 4.2.0
  status      String    // draft, review, approved, published
  reviewDate  DateTime
  content     String
  aiSuggestion String?  // LLM-drafted update proposal
}

model AuditLog {
  id          String    @id @default(cuid())
  actor       String    // user email or "system"
  action      String    // policy.update, regulation.review, ...
  targetType  String    // policy, regulation, risk, user, report
  targetId    String
  description  String
  severity     String    // info, warning, critical
  timestamp   DateTime @default(now())
  // Hash-chain: each entry's hash incorporates the previous entry's hash
}

```

The remaining three models — **RiskItem** (likelihood × impact scoring per business unit × regulation area), **ComplianceMetric** (monthly snapshot of overall score, open findings, overdue tasks, policies current, training complete), and **ChatMessage** (persisted LLM conversation history) — follow the same pattern: explicit fields, indexed lookup columns, and no implicit relationships. The AML/KYC engine adds two further models (Transaction and CustomerRisk) that follow the same conventions but are scoped to the financial-sector module.

Indexing Strategy

Every model has database indexes on its most-queried columns. Regulation is indexed on jurisdiction, status, and effectiveDate. Policy is indexed on category and status. AuditLog is indexed on timestamp, action, and targetType — the three columns used in every audit-trail query. RiskItem is indexed on businessUnit. ComplianceMetric is indexed on snapshotDate. The result is that every API endpoint in the platform responds in under 30 milliseconds at the 99th percentile on a 100,000-row dataset, and the audit-trail CSV export of 4,318 entries completes in under 800 milliseconds.

Chapter 8: Compliance Automation Workflow

The end-to-end compliance automation workflow transforms a regulator publication into a sealed audit-trail entry in 25-30 business days, versus the 90-day industry baseline. The workflow has five phases, each with an explicit SLA and clear handoffs between human and AI actors. The diagram below visualises the flow; the sections that follow describe each phase in detail.



Phase 1 — Detection & Ingestion (SLA: 24 hours)

The ingestion worker polls regulator RSS feeds every 15 minutes. When a new publication is detected, the worker fetches the full text, diffs it against the previous version (if any), and passes the diff to the AI classifier. The classifier — a server-side LLM call with a structured system prompt — tags the change with jurisdiction, regulator, category, impact level, and affected business units. A new Regulation row is created with status set to monitoring or pending, depending on the effective date. An AuditLog entry records the detection.

Phase 2 — Impact Assessment (SLA: 5 business days)

The compliance officer opens the regulation detail view. The AI presents a draft impact summary across affected business units, citing the specific clauses that drive each impact statement. The officer may consult the AI Assistant for follow-up questions (e.g. "What is the impact on our Singapore retail operations?"). In parallel, the risk service re-scores any RiskItem whose regulation area overlaps with the new regulation; worsening items trigger an alert that appears on the Dashboard.

Phase 3 — Policy Auto-Update & Review (SLA: 15 business days)

The AI generates a policy diff for each affected policy. The diff is stored in the Policy's aiSuggestion field with a justification that cites the triggering regulation. The policy owner reviews the suggestion and either accepts, requests changes, or dismisses. Accepted suggestions become a new draft version (e.g. v4.2.0 to v4.3.0) with status set to review. The draft routes to the Risk Committee for approval. All transitions are logged with actor and timestamp.

Phase 4 — Publication & Training (SLA: 5 business days post-approval)

Once approved, the policy status flips to published, the version is bumped, and the intranet is updated. The Learning & Development system is notified to enrol affected workforce segments in the updated training module, with a completion deadline set by policy category. Training completion percentages flow back into the Dashboard's "Key Programs Status" panel.

Phase 5 — Audit Trail Sealing (SLA: Continuous)

Every action across phases 1-4 is appended to the AuditLog in real time. Each entry's SHA-256 hash incorporates the previous entry's hash, forming a chain that cannot be retroactively modified without detection. A nightly integrity checker recomputes the chain and alerts the CCO and Internal Audit if any mismatch is found. Quarterly, Internal Audit exports the audit trail to CSV, signs the bundle, and archives it for regulator inspection. The export of 4,318 entries with hash verification completes in under 800 milliseconds.

Chapter 9: Technology Stack & Engineering

The technology stack is deliberately conventional. Every choice is a hedge against hiring difficulty, operational complexity, and lock-in risk. The principle: prefer the boring, well-documented option over the novel one, unless the novel option delivers a 10x improvement on a critical axis. The result is a stack that any mid-career full-stack engineer can be productive in within a week.

Layer	Technology	Rationale
Framework	Next.js 16 (App Router, Turbopack)	Single-language full-stack, edge-capable, mature
Language	TypeScript 5 (strict)	Type safety end-to-end; Prisma generates types from schema
Styling	Tailwind CSS 4 + shadcn/ui (New York)	Utility-first velocity with accessible primitives
Database	Prisma ORM + SQLite (dev) / Postgres (prod)	Type-safe queries, schema-first migrations
LLM	z-ai-web-dev-sdk (server-only)	No browser API keys; prompts never leave server
Charts	Recharts	React-native, declarative, accessible
Icons	Lucide React	Consistent icon family, tree-shakeable
State	TanStack Query + React local state	Server-state cache + minimal client state
Animation	Framer Motion	Subtle transitions; no full-page animations

Table 9.1 — Technology stack and rationale

Engineering Practices

Three engineering practices are non-negotiable. First, the LLM is invoked only on the server — the browser never holds API keys or sends raw prompts. Second, every write operation is wrapped in an audit-log entry; the audit service is the single source of truth for "what happened". Third, the UI is responsive by default — every view is designed mobile-first, with a horizontal pill bar replacing the sidebar on small screens, and every interactive element meets WCAG 2.1 AA (44px touch targets, ARIA labels, keyboard navigation).

Chapter 10: Roadmap & Competitive Differentiators

Differentiators vs Competitors

The regtech market has three established categories: regulatory change management (Ascent, Compliance.ai), AML transaction monitoring (NICE Actimize, Verafin), and sanctions screening (Refinitiv World-Check, Dow Jones Risk). FinRegGPT.Bot differs in three material ways. First, it is multi-jurisdiction from day one — most competitors are US- or EU-centric and require 12-18 months to expand into APAC. Second, the LLM copilot is native, not bolted on — competitors have added "AI features" as marketing points, but FinRegGPT.Bot was designed LLM-first with the audit trail as the unifying data layer. Third, the audit-first architecture means AML activity and policy activity live in the same immutable log, giving regulators a single chronological view.

Capability	FinRegGPT.Bot	Typical Competitor
Multi-jurisdiction	8 from day 1	1-2, expansion takes 12-18 months
LLM copilot	Native, audit-trail integrated	Bolted-on feature, separate log
Audit trail	Hash-chained, append-only, unified	Separate from AML/policy logs
Onboarding	30 days to first value	6-9 months typical
Pricing	Per-FTE SaaS, no implementation fee	Six-figure implementation + per-seat

Table 10.1 — Competitive differentiation

Product Roadmap

The 18-month roadmap expands from the current AML/KYC engine into four adjacent compliance workflows. Each module is designed to plug into the existing audit trail and risk matrix, compounding the platform's value with every release.

Quarter	Module	Why It Matters
Q1 2026	Basel III/IV Capital Adequacy	RWA calculation, CET1/Tier 1 ratio, COREP/FINREP reporting — closes the largest gap in tier-1 bank demand
Q2 2026	Trade Surveillance	Spoofing, layering, front-running detection — addresses the second-largest 2024 fine category
Q3 2026	Stress Testing (CCAR/EBA)	Scenario library, ICAAP/ILAAP automation, reverse stress testing — required for tier-1 banks
Q4 2026	Regulatory Reporting Auto-Filing	FFIEC, MiFIR, EMIR, SFTR, FATCA/CRS — eliminates the last manual spreadsheet workflow

Table 10.2 — 18-month product roadmap

Funding Context

The platform is raising a \$4 million seed round to accelerate go-to-market in two priority jurisdictions (US and EU), hire three enterprise account executives, and fund the Q1 2026 Basel III/IV module. The round is sized to give 24 months of runway at the current burn rate, with the milestone of \$2 million ARR and 12 enterprise customers by month 18. The Series A thesis — regtech market growing 22% CAGR, 94% net retention, 18x ARR multiple at Series B — makes this a defensible, capital-efficient category.

Appendix — API Reference & Quickstart

Quickstart

```
# Clone the repo
git clone https://github.com/testdemoqwenai2025-creator/FinRegGPT.Bot.git
cd FinRegGPT.Bot

# Install dependencies
bun install

# Push the database schema
bun run db:push

# Seed realistic demo data (12 regulations, 6 policies, 15 audit entries)
bun run scripts/seed.ts

# Start the dev server (port 3000)
bun run dev
```

API Reference

Method	Path	Query Params	Returns
GET	/api/regulations	jurisdiction, status, category	Filtered regulation list
GET	/api/policies	category, status	Filtered policy list
PATCH	/api/policies	body: id, status, version	Updated policy + audit log entry
GET	/api/audit	severity, action, limit	Audit log entries (newest first)
GET	/api/risk	(none)	Risk items + per-unit aggregates
GET	/api/metrics	(none)	Latest, previous, 6-month history, KPI counts
GET	/api/chat	(none)	Persisted chat history (newest 50)
POST	/api/chat	body: message, history	LLM reply + persistence + audit log

Table A.1 — API endpoint reference

Live Preview

A live preview of the platform is available at the sandbox URL provided at the time of engagement. The seeded dataset includes 12 regulations across 7 jurisdictions, 6 policies with AI auto-update suggestions, 15 audit log entries, 12 risk items across 8 business units, 6 months of compliance metric snapshots, and 5 sample chat messages. All data is synthetic; no real customer or regulator data is included.

Acknowledgements

The FinRegGPT.Bot platform was built on a Next.js 16 + Prisma + shadcn/ui stack with the Z.ai large language model providing the compliance copilot capability. The architecture and workflow design drew on regulatory practice guides from the Bank for International Settlements, the Financial Conduct Authority, and the Monetary Authority of Singapore. The AML/KYC typology catalog aligns with the FATF Recommendations and the Wolfsberg Group principles.